Whole Blood Smear XAI Analysis

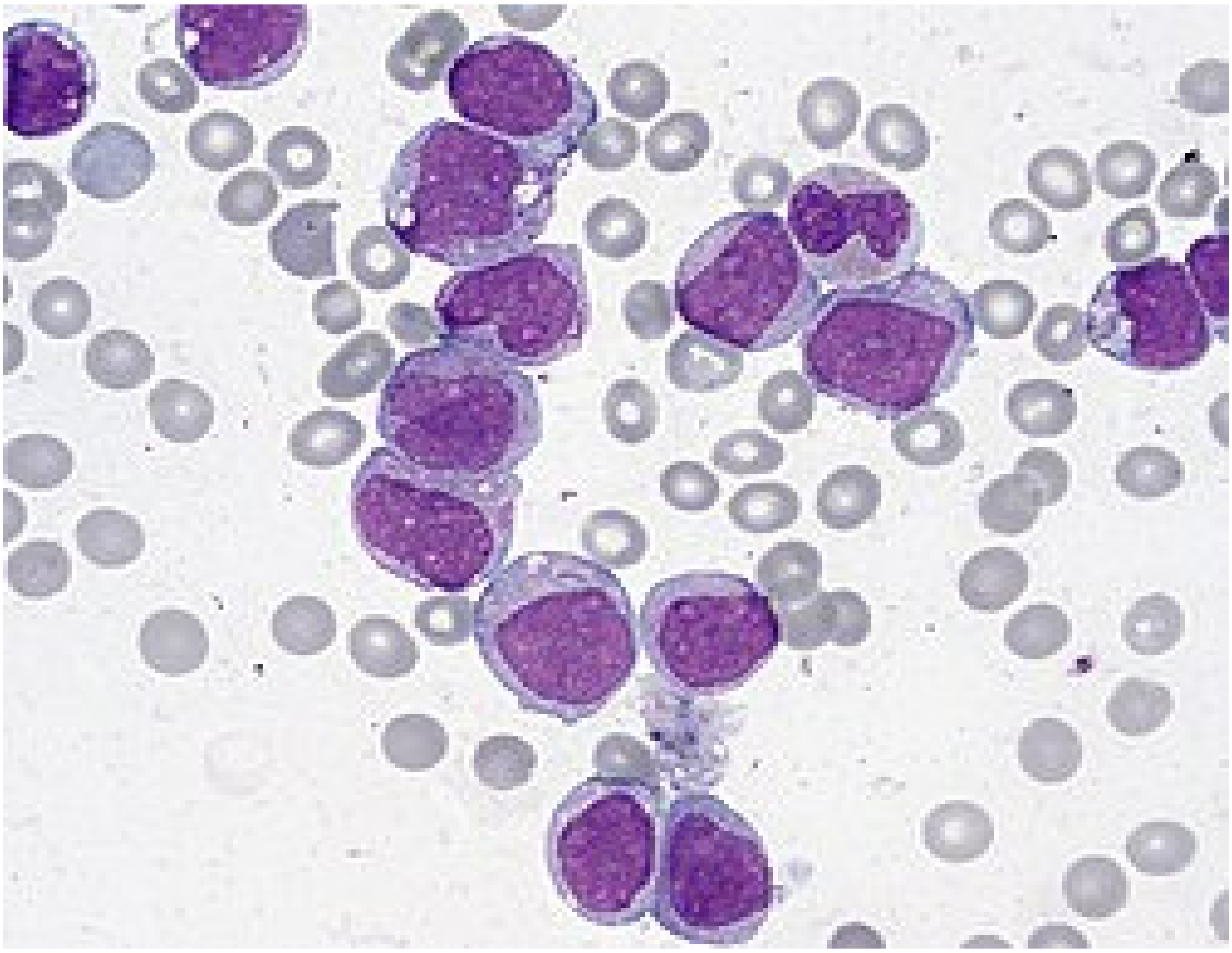
Objective: Research-oriented screening of segmented blood-smear cells for myeloblast-like predictions.

Pipeline: Cellpose segmentation → candidate filtering → ResNet50 classification → Grad-CAM → SHAP.

Measurement	Result
Cellpose objects	83
Nucleated-cell candidates	16
Myeloblast-like candidates	9
Myeloblast-like proportion	56.25%

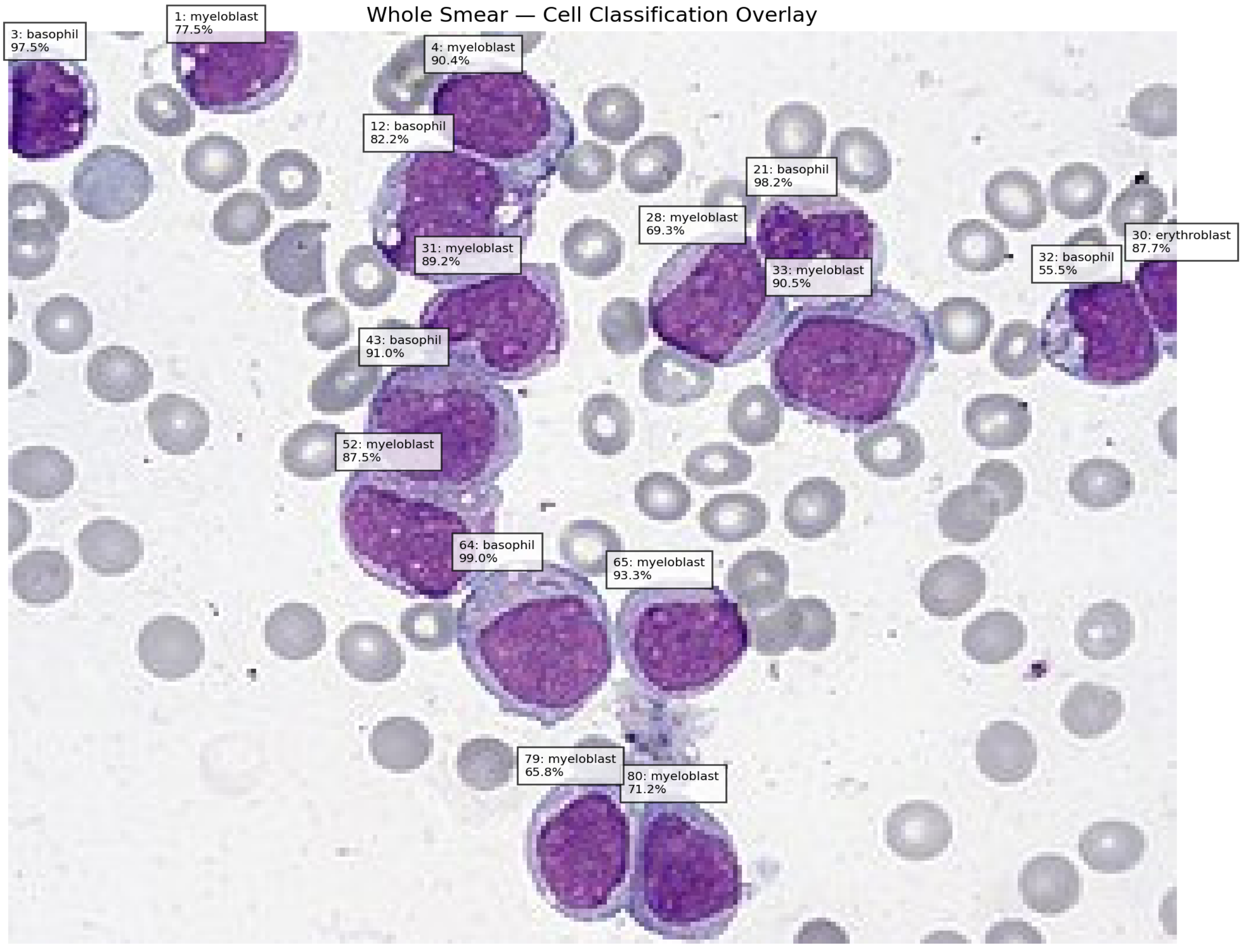
Clinical limitation: This system is a research screening pipeline and is not a clinical diagnostic tool. A myeloblast-like prediction does not establish AML or another hematologic diagnosis.

Whole Smear and Classification



Original input smear used for the analysis.

Classification Overlay



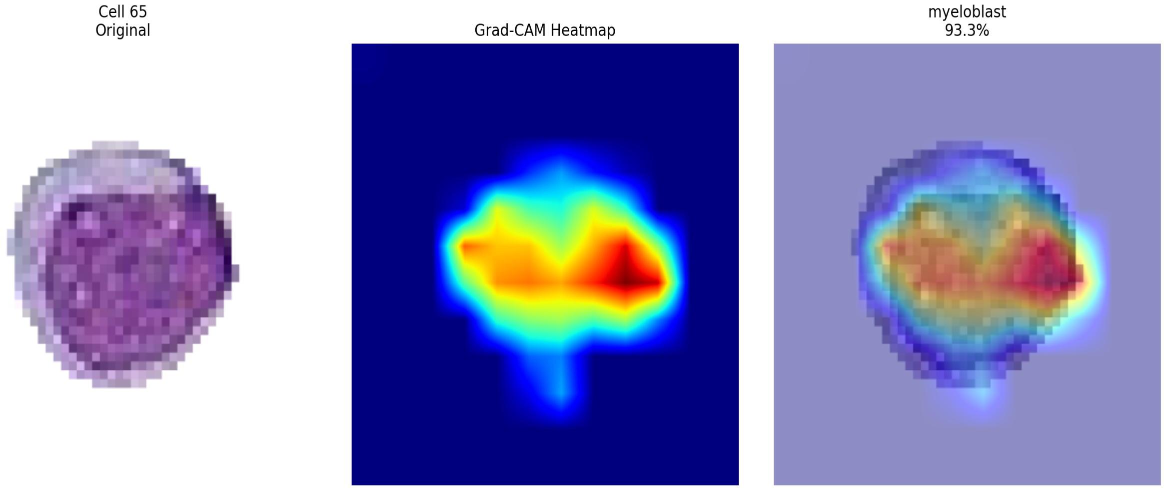
Cell-Level Results

Cell ID	Area	Purple score	Prediction	Confidence
1	446	53.5	myeloblast	77.5%
3	401	57.1	basophil	97.5%
4	598	52.0	myeloblast	90.4%
12	899	47.0	basophil	82.2%
21	607	42.3	basophil	98.2%
28	697	49.1	myeloblast	69.3%
30	149	60.2	erythroblast	87.7%
31	647	52.8	myeloblast	89.2%
32	500	51.5	basophil	55.5%
33	895	47.4	myeloblast	90.5%
43	795	49.2	basophil	91.0%
52	822	48.8	myeloblast	87.5%
64	989	43.8	basophil	98.9%
65	631	50.2	myeloblast	93.3%
79	658	49.2	myeloblast	65.8%
80	689	51.1	myeloblast	71.2%

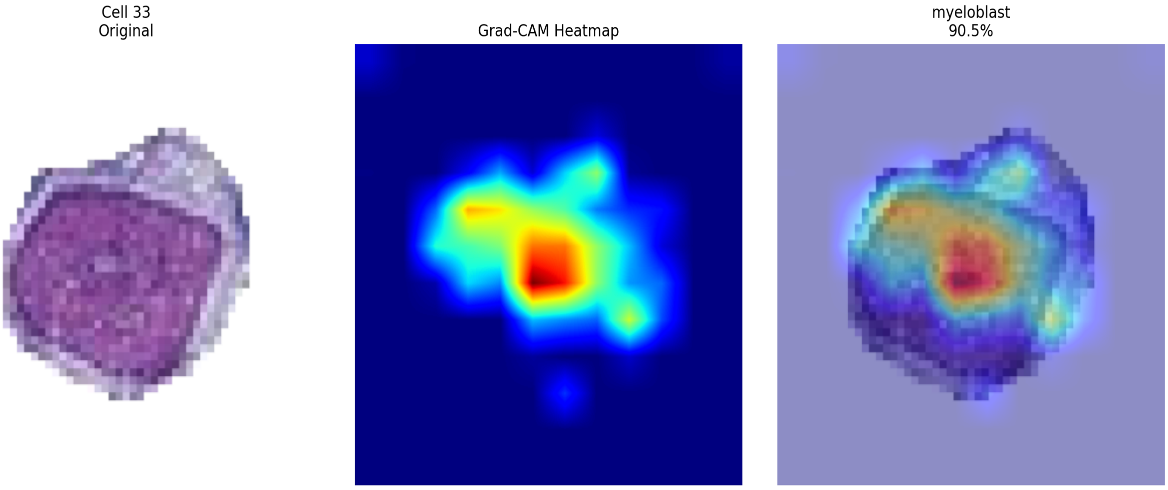
Grad-CAM Explanations

Grad-CAM highlights spatial regions that contributed to the selected model prediction. The heatmap is an explanation of model behavior and should not be interpreted as proof of a specific biological structure.

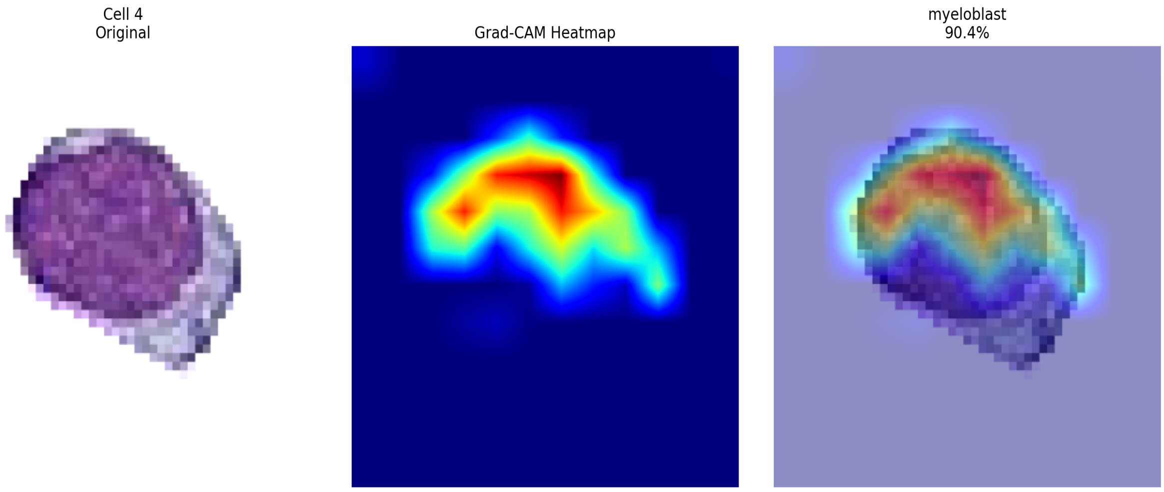
Cell 65 — myeloblast (93.3%)



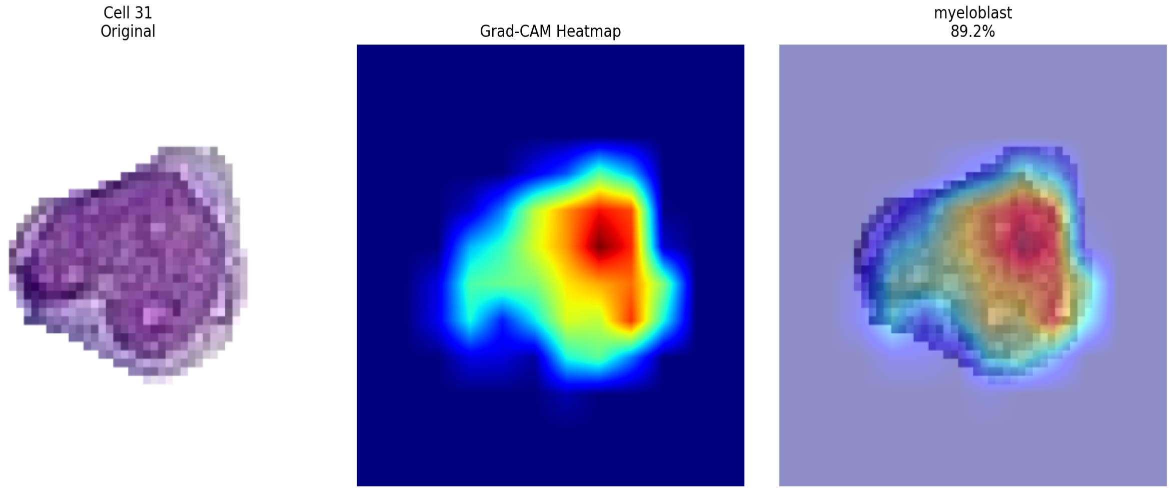
Cell 33 — myeloblast (90.5%)



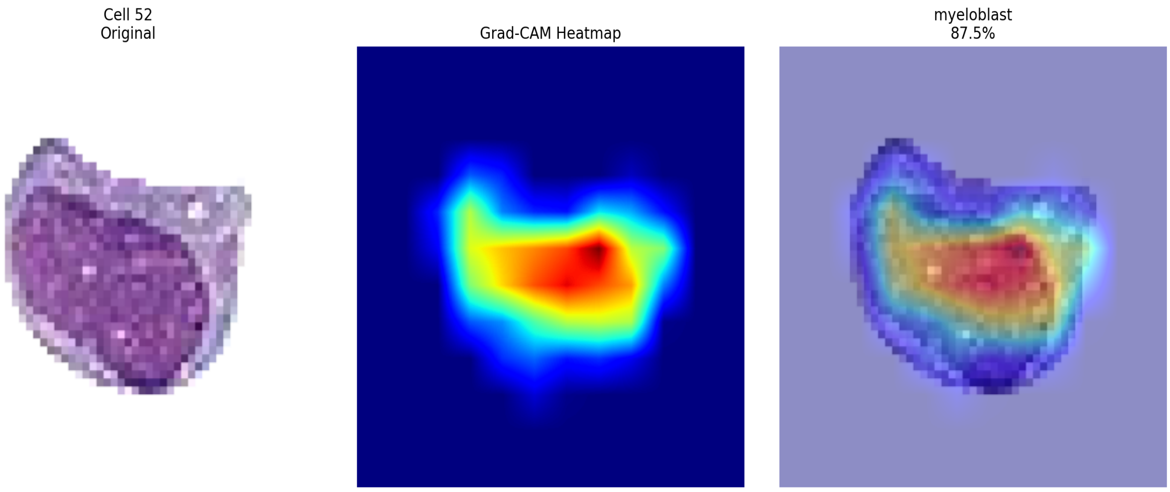
Cell 4 — myeloblast (90.4%)



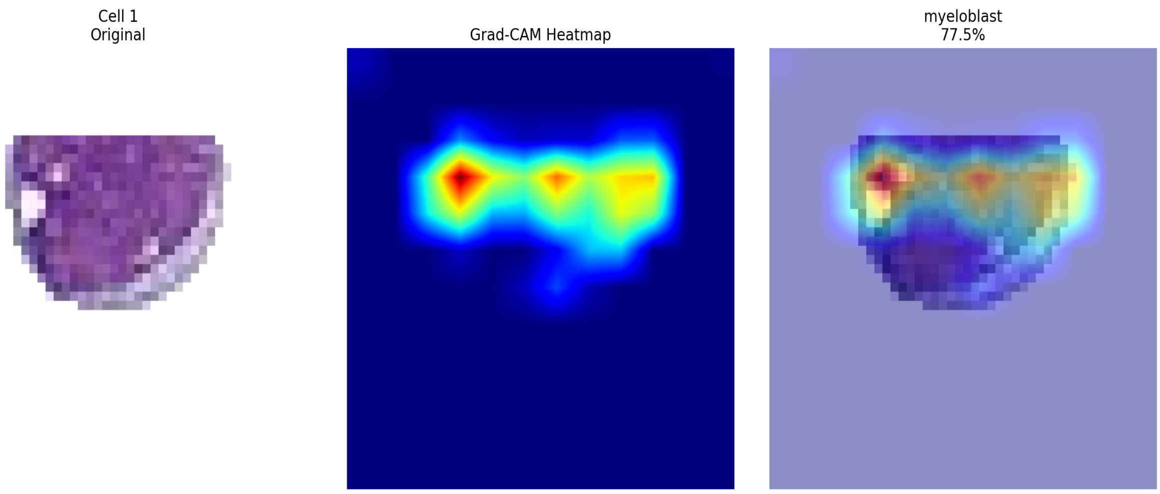
Cell 31 — myeloblast (89.2%)



Cell 52 — myeloblast (87.5%)



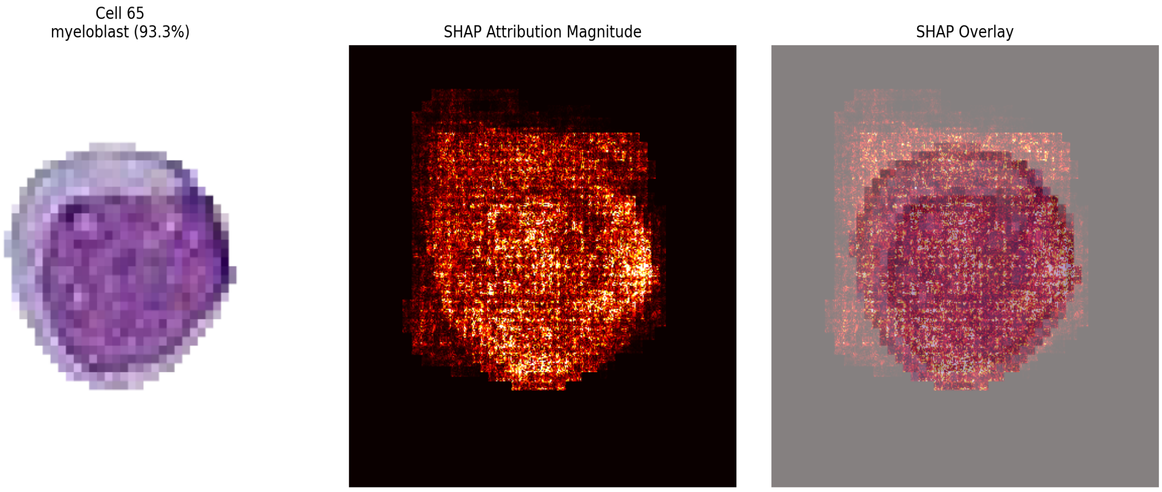
Cell 1 — myeloblast (77.5%)



SHAP Explanations

SHAP attribution magnitude represents the magnitude of input-pixel contribution toward the selected prediction. The visualizations are model explanations rather than direct morphological measurements.

Cell 65 — myeloblast (93.3%)



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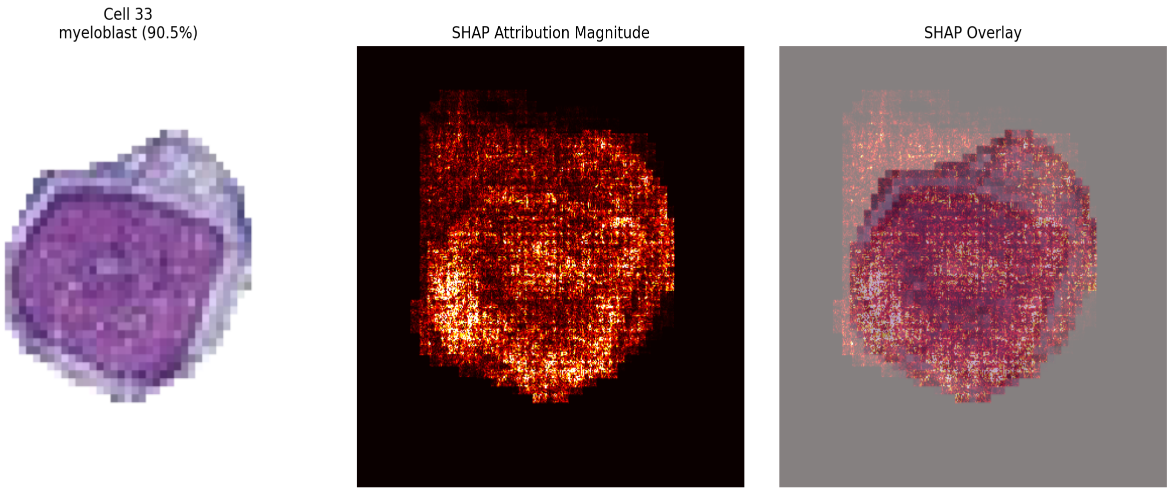
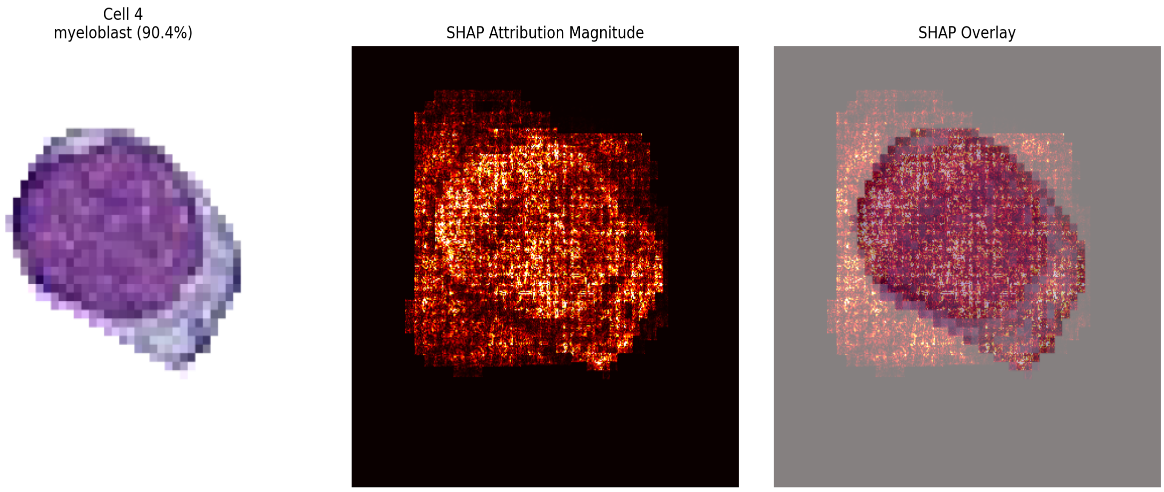


Cell 4 — myeloblast (90.4%)



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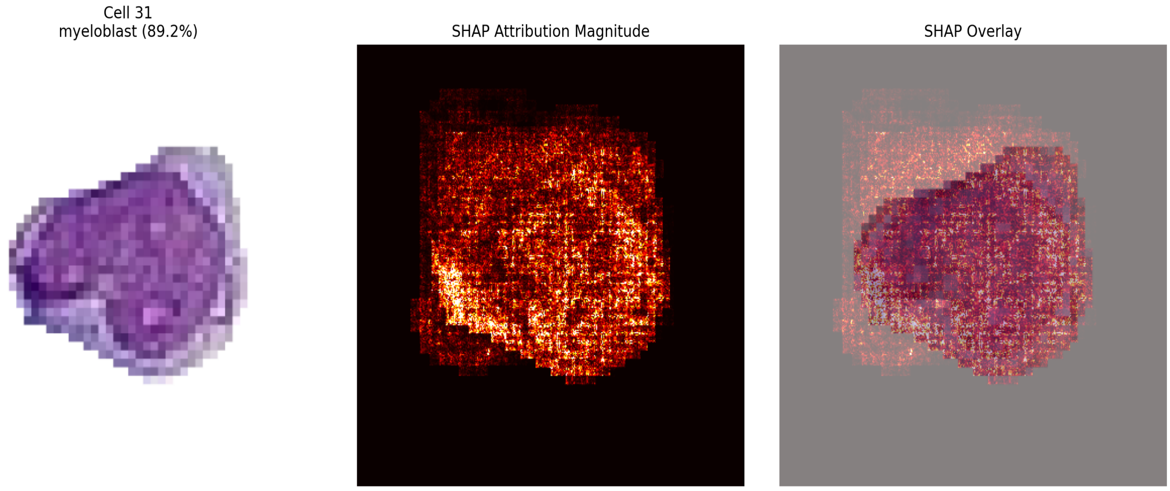

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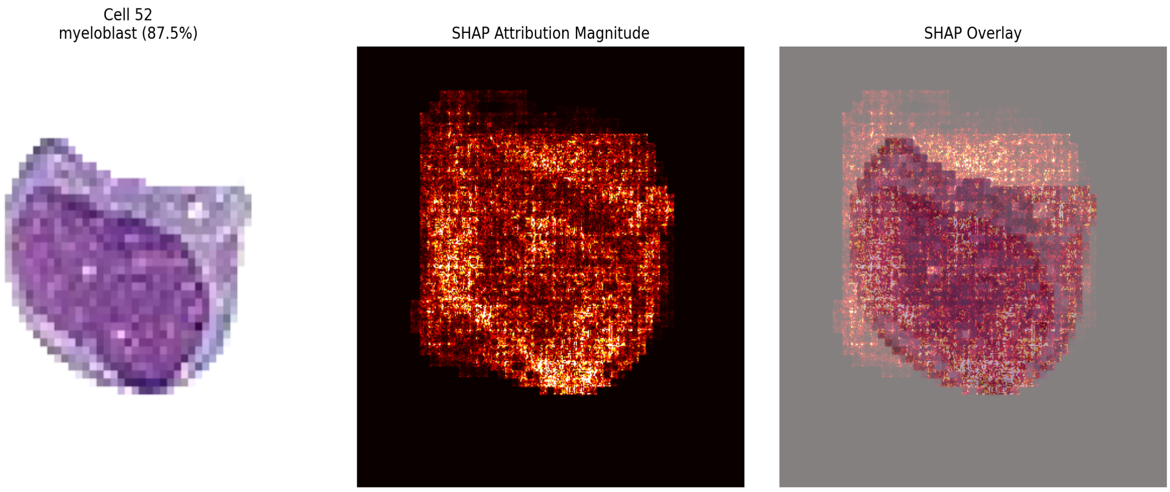

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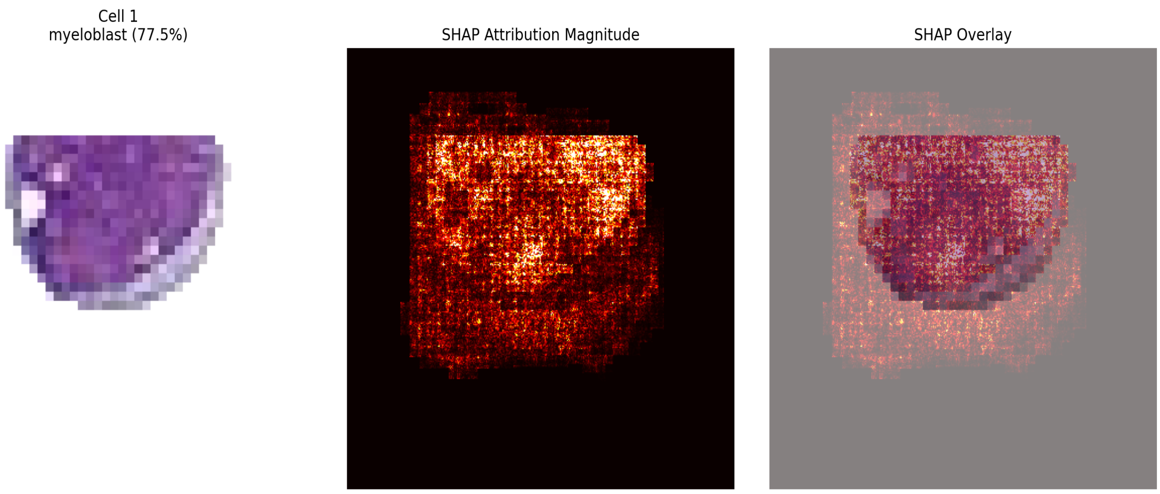

Limitations

• The ResNet50 model was trained using five cell classes.

• RBCs are not an explicit class in the current model and may therefore be assigned incorrectly to one of the available classes.

• Cellpose segmentation errors can affect downstream classification.

• The external smear may differ from the training domain in staining, illumination, acquisition and morphology.

• Model confidence does not guarantee prediction correctness.

• Myeloblast-like predictions should not be interpreted as AML diagnosis.

• Clinical diagnosis requires appropriate hematopathological and laboratory assessment.